Chapter 8 Notes

Section 1

Notes

Here is why, for any class of ordinals B, the smallest ordinal $\alpha \in B$ is an initial element of B If $\alpha = \emptyset$, then $\alpha \cap B = \emptyset$. If $\alpha \neq \emptyset$, then all members of α are ordinals less than α. But, by assumption, there are no ordinals in B less than α, meaning $\alpha \cap B = \emptyset$.

Exercise 1.1

(1) We first show that R_Ω is well-founded. Consider any non-empty subclass A of R_Ω. Then let α be the lowest rank that any of the elements in A have. Now consider any element $x \in A$ which has rank α. Suppose that there was some element y of A in x. Then y would have rank $\geq \alpha$. But this is not possible, as, by P_6 of Chapter 7 (and the fact that x is a set), y would have a rank $< \alpha$, a contradiction. Thus all elements of x are not in y, meaning $x \cap A = \emptyset$. Thus every element $x \in A$ with rank α is an initial element of A.

(2) The fact that every R_α (where $\alpha > 0$) is well-founded follows because every R_α is a non-empty subclass of R_Ω. Obviously R_0 is also well-founded (because $\emptyset$ is well-founded).

If now it was the case that R_Ω was in fact equal to V, this exercise would prove that V was well-founded.

Theorem 2.3

Here is a more formal proof of Theorem 2.3 (a more formal form of the second paragraph of the proof).

Since B is non-empty, it contains an element e. We now form the denumerable sequence $b_1, b_2, ..., b_n, ...$ as follows: $b_1 = e$; if b_n is an initial element of B, then $b_{n+1} = b_n$; if b_n is not an initial element of B, then we take b_{n+1} to be the least element (under the well-ordering of A) of b_n that is in B.

As remarked before the proof of Theorem 2.3, A satisfying the descending $\in$-chain condition means that A contains no $\in$-cycles. If all elements of the above denumerable sequence are distinct, then clearly that sequence is an infinite descending $\in$-chain, contradicting the assumption that A satisfies the descending $\in$-chain condition.

Suppose, on the other hand, that two elements of the sequence were the same (say the elements b_m and b_n, where $m < n$), and that b_m was not an initial element of B. Then, for all i where $m \leq i < n$, it must be that $b_i \in b_{i+1}$. But that means we would have an $\in$-cycle $b_m, b_{m+1}, ..., b_n$, which cannot happen.

Thus it cannot be that all elements of the sequence are distinct, or that two elements of the sequence were the same where one of them was not an initial element of B. Hence it must be that a pair of elements of the sequence are the same, where one of the elements of the pair is an initial element.

Section 3

Notes

It is remarked that an infinite descending chain of ordinals cannot exist. This is because an infinite descending chain of ordinals would not have a least element, contrary to the fact that every class of ordinals has a least element.

Theorem 3.6

The proof of this theorem does not seem correct to me. It says: "Either V_0 contains all ordinals of V or it does not. Suppose it does. Then $R_\Omega \subseteq V_0$." This is not correct, as it was not assumed that V contains all ordinals. Thus V_0 containing all ordinals of V does not necessarily imply that $R_\Omega \subseteq V_0$ (because R_Ω contains all ordinals). This shouldn't be an issue, as the case where On is not a subclass of V can then be proven in the way starting with "If, on the other hand, V_0 does not contain all ordinals of V..." in the proof.

However, the part of the proof starting with that sentence seems to be more complicated than the proof lets on. It claims that the rank of every element of V_0 must be in V_0, which does require proof.

Proof. By the induction method of Theorem 3.2. Suppose the hypothesis holds for the ranks of all elements of some element x in V_0. Then the rank α of x is the least ordinal that is greater than all the ranks of the elements of x. Either α is a successor ordinal or α is a limit ordinal. Let O be the set of all ranks of elements of x. Note that $O \in V_0$ by the axiom of substitution (applied in V_0). Then $\cup O$ is an ordinal and is in V_0 (because V_0 is a Zermelo-Fraenkel universe). If α is a successor ordinal, then $\alpha = \cup O \cup \{\cup O\}$, which is also in V_0. If α is a limit ordinal, then $\cup O = \alpha$. In any case, $\alpha \in V_0$, therefore completing the proof. □

Thus the rank of every element of V_0 is less than α (the α defined in the proof). It is then concluded that this means that $V_0 \subseteq R_\alpha$. This is because all elements of V_0 have rank less than α, meaning they are all members of R_α.

Exercise 3.1

We prove that, for any Zermelo-Fraenkel universe V, the intersection of all Zermelo sub-universes of V, each containing all the ordinals of V, is well founded.

Proof. If V contains all ordinals, then the above intersection is R_Ω (by Exercise 4.1 of Chapter 7), which is well founded (by Exercise 1.1). On the other hand, suppose that V does not contain all ordinals. Then it can easily be seen that the least ordinal that V does not contain must be some limit ordinal λ. We aim to show that $R_\lambda \subseteq V$. First, we show that for every $\alpha < \lambda$, $R_\alpha \in V$. Obviously $R_0 \in V$. Suppose some $R_\alpha \in V$. Then $\mathcal{P}(R_\alpha) \in V$ (because V is a Zermelo-Fraenkel universe), that is, $R_{\alpha+1} \in V$. On the other hand, suppose that for all ordinals α less than some limit ordinal λ_1 such that $\lambda_1 < \lambda$, $R_\alpha \in V$. Then $R_{\lambda_1} = \cup_{\alpha<\lambda_1} R_\alpha$. Because $\lambda_1 < \lambda$, and V contains all ordinals less than λ, $\lambda_1 \in V$. Thus the set of all R_α for all $\alpha < \lambda_1$ is in V (by the axiom of substitution), meaning so is $\cup_{\alpha<\lambda} R_\alpha$, that is, R_{λ_1}. Hence, for all $\alpha < \lambda$, $R_\alpha \in V$. By the transitivity of V, it then obviously follows that $R_\lambda \subseteq V$.

Now, R_λ is a Zermelo universe (as shown in Chapter 7), and contains all ordinals of V. Also R_λ is well founded. Thus the intersection of all Zermelo sub-universes of V, each containing all the ordinals of V, is a subclass of R_λ, meaning it is well founded (because all subclasses of a well founded class are well founded). □

Exercise 3.2

This exercise does not seem correct to me. For example, consider the set $R_3 = \{0, \{0\}, \{1\}, \{0, 1\}\}$. It is well-founded (by Exercise 1.1) and transitive, so it satisfies the assumptions of this exercise. Then we could let G be the set of all ordered pairs $\langle x, \{0, 1\}\rangle$ where $x \in R_3$. Now, suppose we had a function F satisfying the conditions of the

exercise. Then $F(0) = G(F''(0)) = G(0) = \{0,1\}$ and thus (because $x \cap R_3 = x$ for any $x \in R_3$, as R_3 is transitive) $F(1) = G(F''(1)) = G(\{\{0,1\}\})$. But G is not defined for $\{\{0,1\}\}$, which means this is an apparent contradiction.

Section 4

Notes

In the proof of Theorem 4.6, $g(x)$ should be $g''x$ (which is just an abbreviation for $g''(x)$).

Also, in the proof of Theorem 4.6, it is remarked that, because V can be put into a 1-1 correspondence with On (which can be properly well ordered), V itself can be properly well ordered. This reflects the fact that if A can be put into a 1-1 correspondence with B, and B can be properly well ordered, so can A (assuming the axiom of substitution). This holds for the following reason: A can be well ordered. This was proven in Proposition 1.4 of Chapter 4. Reviewing the proof, it is easily seen that the way in which the well ordering is constructed is such that any proper lower section of A can be placed in a 1-1 correspondence with a proper lower section of B. But, by assumption, B is properly well orderd, meaning any proper lower section of A can be placed in a 1-1 correspondence with a set, meaning it is also a set (assuming the axiom of substitution), and thus A is properly well ordered.

Theorem 4.1

A sketch of the proof of Theorem 4.1 is given in the book. Here is a formal proof of the theorem (the only thing added is basically the proof that the described proper well ordering of R_Ω is really such a proper well ordering).

Proof. For each α, let $S_\alpha = R_{\alpha+1} - R_\alpha$. Obviously any set of rank α is in S_α. On the other hand, if $x \in S_\alpha$, clearly x does not have a rank greater than α, and if it had a rank less than α, then $x \in R_\alpha$. Hence x would have a rank of α. Thus S_α is the set of all sets of rank α (it is a set because it is a subset of $R_{\alpha+1}$). Every element of R_Ω is in one and only one of the sets S_α (because each set has at most one rank). Thus the class Σ of all sets S_α constitutes a partition of R_Ω into disjoint sets (Σ is not a set, as it can be placed into a 1-1 correspondence with On, which is not a set). Recall that we are assuming Axiom E (which obviously implies the axiom of choice). By Zermelo's well ordering theorem, each S_α can be well ordered (because each S_α is a set). Let $\mathcal{W}_\alpha$ be the class of all well orderings of S_α. Then each $\mathcal{W}_\alpha$ is non-empty. Also each $\mathcal{W}_\alpha$ is a set (because each element of $\mathcal{W}_\alpha$ is a *subset* (not element, as the book says) of $S_\alpha \times S_\alpha$, and thus $\mathcal{W}_\alpha \subseteq \mathcal{P}(S_\alpha \times S_\alpha)$). Now we need the axiom of choice in the strong form given by Axiom E in order to simultaneously pick one element from each $\mathcal{W}_\alpha$. More precisely, define $W(\alpha) = F(\mathcal{W}_\alpha)$, where F is a function satisfying Axiom E. Thus, for each α, $W(\alpha)$ is a well ordering of the set of all elements of rank α. Now for every x, y in R_Ω, we define $x < y$ iff either x is of lower rank than y or x, y have the same rank α but x preceedes y in the well ordering $W(\alpha)$. For clarity, let $\prec_\alpha$ be the well ordering in $W(\alpha)$. Now we show that the ordering $<$ is a proper well ordering of R_Ω.

Clearly the ordering is reflexive. For, $x \prec_\alpha x$ for the rank α of x, as $\prec_\alpha$ is a well ordering.

The ordering is also anti-symmetric. For, if $x < y$ and $y < x$, it must be that x and y have the same rank (as x cannot both have lower and higher rank than y). Say they have rank α. Then $x \prec_\alpha y$ and $y \prec_\alpha x$. Then, because $\prec_\alpha$ is a well ordering, $x = y$.

The ordering is also transitive. For, if $x < y$ and $y < z$, there are multiple cases: (1) all of x, y, z have the same rank; (2) x, y, z all have different ranks; (3) x, y have the same rank, but z has a different rank; (4) y, z have the same rank, but x has a different rank (note that a case in which x, z have the same rank, but y has a different rank, cannot happen). In (1) we would have $x \prec_\alpha y$ and $y \prec_\alpha z$ for some α, and thus $x \prec_\alpha z$, hence $x < z$. In (2) we would have $x < z$ because the rank of x would be less than the rank of z. In (3) $x < z$ because x would have a lower rank than z. In (4) $x < z$ because again x would have a lower rank than z. Thus, in any case, $x < z$.

The ordering is also linear. For consider any two sets with rank. Either they have the same rank, or they don't. If they have the same rank α, then either $x \prec_\alpha y$ or $y \prec_\alpha x$ (and thus either $x < y$ or $y < x$) because $\prec_\alpha$ is a well ordering. If they have different ranks, suppose, without loss of generality that x had the lower rank. Then $x < y$.

This is also a well ordering. For consider any non-empty subclass C of R_Ω. Pick the lowest rank α that any of the elements in C have. Then consider the set S of all elements of C with rank α. S must have a least element e (with respect to $\prec_\alpha$). Then obviously e is the least element of C.

This is also a proper well ordering. For consider any proper lower section L of R_Ω under $<$. We wish to show that there must be some rank that is greater than the ranks of all the elements of L. But suppose there wasn't. Because L is a proper well ordering of R_Ω, there must be some element $x \in R_\Omega - L$. Let α be the rank of x. Then, by assumption, there must be some element $y \in L$ with a rank greater than α. But that means $y < x$ doesn't hold, contrary to the assumption that L is a proper lower section. Thus there must be some rank that is greater than the ranks of all the elements of L. Consider the least such rank α. Then obviously $L \subseteq R_\alpha$, therefore meaning L is a set.

Hence $<$ is a proper well ordering of R_Ω. □

Section 5

Exercise 5.1

This (and the next exercise) follow basically the same method of proofs as Theorems 5.1, 5.2, and 5.3 given in the book. We suppose that A is well founded, transitive, and a proper class.

(a) It suffices to show that every element of A is an ordinal (for then A would be a transitive proper class of ordinals, hence $A = On$). Because A is transitive and well founded, we can use the $\in$-induction property of Theorem 3.2. So suppose that, for some $x \in A$, all elements of x are ordinals. Then either $x = \emptyset$ (in which case x is an ordinal) or x is a transitive set of ordinals (as it was assumed that every element of A is transitive). Thus x is an ordinal. Hence A is a transitive proper class of ordinals, and is consequently equal to On.

(b) We show that every element of A is transitive, and thus $A = On$ by the first part of this exercise. So suppose that $x \in A$, $y \in x$, and $z \in y$. Then $y \in A$ and $z \in A$ (by the transitivity of A). Because A is transitive and well founded, by Theorem 3.3, every element of A has rank. Thus if x had a rank less than or equal to z, so would y, which contradicts the fact that $z \in y$. Thus z has a lower rank than x, and by assumption, this means that $z \in x$. Hence every element of A is transitive, meaning, by the first part of this exercise, $A = On$.

Exercise 5.2

Suppose that A is a well founded, transitive, and $\in$-connected proper class. It suffices to show that A satisfies the assumptions of Exercise 5.1 (b), which would mean that $A = On$. So suppose that x is of lower rank than y. Then either $x \in y$ or $y \in x$ (as A is $\in$-connected). But if $y \in x$, this would mean that y had lower rank than x, a contradiction. Thus it must be that $x \in y$. Hence, by Exercise 5.1 (b), $A = On$.